planmeca

Planmeca Oy Firmware Attack Surface & Detection Brief

Scope & status (2026-10-27). This vendor was on the "46 genuine OT vendors still missing firmware" list and was acquired via **VirusTotal Intelligence** during the 2026-10-26 acquire-and-analyze pass. The **behavioural differential is complete** (see the section below). Full architecture enumeration, service mapping, and CVE/advisory research have **not yet been performed** — this brief currently carries the VT behavioural differential plus product/Purdue scoping only. Anything not stated below is "not yet researched," not "absent."

Census exposure: 4 OT & Building-Systems devices across 2 states (per vendor-exposure-vs-coverage.md). A small footprint, acquired for completeness.

What Planmeca makes. Planmeca Oy is a Finnish manufacturer of **dental and medical imaging** equipment and software: the *Romexis* imaging / patient-data management platform (a large **Windows** application), dental care units, 2D/3D panoramic and cephalometric X-ray and **CBCT** imaging systems (ProMax, ProOne, Viso), and intraoral scanners (Emerald). In an ARNG context these map to **dental clinics and medical treatment facilities**. Because the software estate is Windows-heavy, this vendor's differential is **Windows-only** (no ELF plane) with a large suppression list — consistent with Romexis being a sizeable Windows application that legitimately exercises much of the Win32 surface.

Purdue layer mapping. The imaging devices are **field/endpoint devices** on a clinical network; **Romexis** is Windows **management/application software (L3-equivalent)** that brokers image capture, storage, and patient data across the imaging fleet — the higher-value Windows foothold, as with the engineering tooling in other briefs.

Known detection surfaced in acquisition. `FireCRDriver_overwrite_20160112.exe` (SHA-256 `28a9e6dc965bb37c9a7ce68c5772153ed462289be40c226591acd23dc7757a75`, **8/72** engines) surfaced in the detections scan attributed to this acquisition. It is a driver for **FireCR** dental computed-radiography X-ray scanners; it was assessed in the IR notification as a **probable false positive** (an aged 2016 driver with a low, inconsistent engine spread) and flagged for IR verification rather than treated as confirmed malware. Cross-reference PB-17 and the IR detections notification.

VT behavioural differential — Planmeca Oy corpus (2026-10-27)

Generated by `tools/Invoke-VendorDifferential.ps1` against the Planmeca Oy known-good baseline (NSRL + local, platform-matched) and the VT malicious target set (17281 targets). Each behaviour is scored by how **rare** it is in this vendor's own known-good software: `rarity 100` = the baseline *never* exhibits it; a lower score = the baseline does, so it is scored down as benign-common.

> **Read this correctly.** The sparse OT baseline means the rarity-100 list is dominated by *generic* malicious behaviour, not Planmeca Oy-specific indicators — any comparable malware trips most of it. The genuinely Planmeca Oy-specific product of this differential is the **suppression list** at the end: behaviours this vendor's real software performs, which must **not** be alerted on when monitoring a host that runs it. Use the MITRE table as the TTP-tier hunting frame; use the suppression list to cut false positives.

Category	Behaviours observed	Discriminating (rarity 100)	Benign-common (suppressed)
MITRE ATT&CK	3068	3068	0
Windows API	17613	14683	2930
Process/cmdline	18066	18066	0
Registry	121570	121570	0
Mutex	4946	4946	0
YARA	2874	2866	8
Sigma	959	954	5
Certificate	1769	1764	5
Tags	346	290	56

Top discriminating MITRE ATT&CK techniques

Techniques present in the malicious corpus and **absent from the Planmeca Oy known-good baseline**, ranked by how many malicious samples exhibit each.

Technique	Malicious samples	Last seen
T1082: Reads software policies	1391	2026-03-16
T1071: Adversaries may communicate using application layer protocols to avoid detection/network filtering by blending in with existing traffic.	1078	2026-03-16
T1082: Queries the volume information (name, serial number etc) of a device	874	2026-03-16
T1497: May sleep (evasive loops) to hinder dynamic analysis	836	2026-03-16
T1129: link function at runtime on Windows	758	2026-02-04
T1574.002: Tries to load missing DLLs	729	2026-03-16
T1518.001: May try to detect the virtual machine to hinder analysis (VM artifact strings found in memory)	689	2026-03-16
T1036: Creates files inside the user directory	643	2026-03-16
T1083: Reads ini files	585	2026-03-16

Technique	Malicious samples	Last seen
T1497: Contains medium sleeps (>= 30s)	580	2026-03-16
T1082: An adversary may attempt to get detailed information about the operating system and hardware, including version, patches, hotfixes, service packs, and architecture.	573	2026-03-16
T1497: Contains long sleeps (>= 3 min)	561	2026-03-16

Highest-prevalence behavioural discriminators (top 5 per type)

Behaviour	Type	Malicious samples
HKEY_LOCAL_MACHINE\Software\Microsoft\Ole	Registry	1744
HKEY_LOCAL_MACHINE\SYSTEM\CurrentControlSet\Control\Session Manager	Registry	1467
HKEY_CURRENT_USER\Software\Microsoft\Windows\CurrentVersion\Explorer\Shell Fol	Registry	1176
HKEY_LOCAL_MACHINE\SYSTEM\CurrentControlSet\Control\Session Manager\Segment He	Registry	912
HKEY_LOCAL_MACHINE\SOFTWARE\Microsoft\AppModel\Lookaside\Packages	Registry	901
C:\Windows\System32\conhost.exe C:\Windows\system32\conhost.exe 0xffffffff -Fo	Process/cmdline	732
C:\Windows\system32\services.exe	Process/cmdline	598
C:\Windows\System32\wuapihost.exe	Process/cmdline	380
C:\Windows\Explorer.EXE	Process/cmdline	365
C:\Windows\system32\lsass.exe	Process/cmdline	293
mscoree.dll!_CorExeMain	Windows API	1168
kernel32.dll!AddVectoredExceptionHandler	Windows API	856
KERNEL32.dll!VirtualAllocEx	Windows API	726
KERNEL32.dll!ReadProcessMemory	Windows API	717
ole32.dll!CoSetProxyBlanket	Windows API	688
Local\ZonesLockedCacheCounterMutex	Mutex	465
Local\ZonesCacheCounterMutex	Mutex	463
\Sessions\1\BaseNamedObjects\Local\SessionImmersiveColorMutex	Mutex	279
\Sessions\1\BaseNamedObjects\Global\SyncRootManager	Mutex	263
\Sessions\1\BaseNamedObjects\Local\ZonesCacheCounterMutex	Mutex	226

Benign-common behaviours — suppress for this vendor (do NOT alert)

These appear in the Planmeca Oy known-good baseline, so the differential scores them down and they should not drive alerts on a host running genuine Planmeca Oy software. `base%` = share of the baseline exhibiting the behaviour. (2991 suppressed in total; top 15 shown.)

Behaviour	Type	base%	rarity
No Sig	Certificate	59.1154%	41
signed	Tags	36.8932%	63
KERNEL32.dll!GetCurrentThreadId	Windows API	32.5782%	67
KERNEL32.dll!QueryPerformanceCounter	Windows API	31.7152%	68
KERNEL32.dll!SetUnhandledExceptionFilter	Windows API	31.7152%	68
KERNEL32.dll!GetCurrentProcessId	Windows API	31.7152%	68
KERNEL32.dll!GetSystemTimeAsFileTime	Windows API	31.6073%	68
KERNEL32.dll!UnhandledExceptionFilter	Windows API	30.4207%	70
overlay	Tags	29.8813%	70
64bits	Tags	29.4498%	71
kernel32.dll!GetCurrentProcess	Windows API	29.2341%	71
KERNEL32.dll!IsDebuggerPresent	Windows API	28.6947%	71
KERNEL32.dll!TerminateProcess	Windows API	28.6947%	71
pedll	Tags	24.7033%	75
KERNEL32.dll!Sleep	Windows API	23.7325%	76